Program 1

Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document.

```
\documentclass{article}
\usepackage{fancyhdr}
\usepackage{lipsum} % for dummy text
\pagestyle{fancy}
\fancyhf{} % Clear header and footer
\fancyhead[R]{Latex Programming} % Right header
\fancyfoot[R]{\thepage} % Right header
\fancyfoot[L]{VVIET, Mysore} % Left footer
\begin{document}
    \section{Section 1}
    \lipsum[1-2]
    \section{Section 2}
    \lipsum[1]
\end{document}
```

Output:

Latex Programming

1 Section 1

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

2 Section 2

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

VVIET, Mysore 1

Program 2

Develop a LaTeX script to create a document that displays the sample Abstract/Summary.

```
\documentclass{article}
\title{Sample Abstract/Summary}
\author{}
\date{}
\begin{document}
\maketitle
\begin{abstract}
This is a sample abstract. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod
tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud
exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.
\end{abstract}
\end{document}
```

Output:

Sample Abstract/Summary

Abstract

This is a sample abstract. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Program 3

Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting.

```
\documentclass{article}
\usepackage{graphicx}
\usepackage{geometry}
\begin{document}
\begin{titlepage}
\centering
\includegraphics[width=0.3\textwidth]{vtulogo.png}\par
\vspace{1cm}
{\scshape\LARGE Visvesvaraya Technological University \par}
\vspace{1cm}
{\scshape\Large Project Report\par}
\vspace{1.5cm}
{\huge\bfseries Title of the Project\par}
\vspace{2cm}
{\Large\itshape Your Name\par}
\vfill
Guide: \textsc{Guide's Name}\par
\vfill
{\large \today\par}
\end{titlepage}
\end{document}
```

Output:

Visvesvaraya Technological University

Project Report

Title of the Project

Your Name

Guide: Guide's Name

June 18, 2025

Program 4

4. Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry]

```
\documentclass{article}
\begin{document}
    \begin{center}
        \textbf{\LARGE Certificate}
    \end{center}
    \vspace{1cm}
    This is to certify that \underline{\hspace{6cm}} has completed the report titled
\underline{\hspace{8cm}} for the course \underline{\hspace{5cm}}.
    \vspace{2cm}
    \begin{flushright}
        Date: \underline{\hspace{3cm}}
        \underline{\hspace{8cm}} \\
        \textit{(Signature of Supervisor)}
    \end{flushright}
\end{document}
```

Output:

Certificate

This is to certify that ______________________________ has completed the report titled ____ for the course _________________________.

Date: _______________ ____

(Signature of Supervisor)

Program 5

Develop a LaTeX script to create a document that contains the following table with proper labels.

SL NO	USN	Student Name	Marks		
			Subject 1	**Subject 2**	**Subject 3**
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

```
\documentclass{article}
\usepackage{multirow}
\begin{document}
    \begin{table}
        \caption{Student Results}
        \centering
        \begin{tabular} {|c|c|c|c|c|c|}
            \hline
            \multirow{2}{*}{SL NO} & \multirow{2}{*}{USN} &
            \multirow{2}{*}{Student Name}&  \multicolumn{3}{|c|}{MARKS}\\
            \cline{4-6}& & &JAVA & DBMS & Python \\
            \hline
            1 & 4VM22IS001 &  ABC & 89 & 60 & 90 \\
            \hline
            2 & 4VM22IS002 &  MNQ & 75 & 45 & 98 \\
            \hline
            3 & 4VM22IS003 &  XYZ & 67 & 55 & 59 \\
            \hline
        \end{tabular}
    \end{table}
\end{document}
```

Output:

Table 1: Student Results

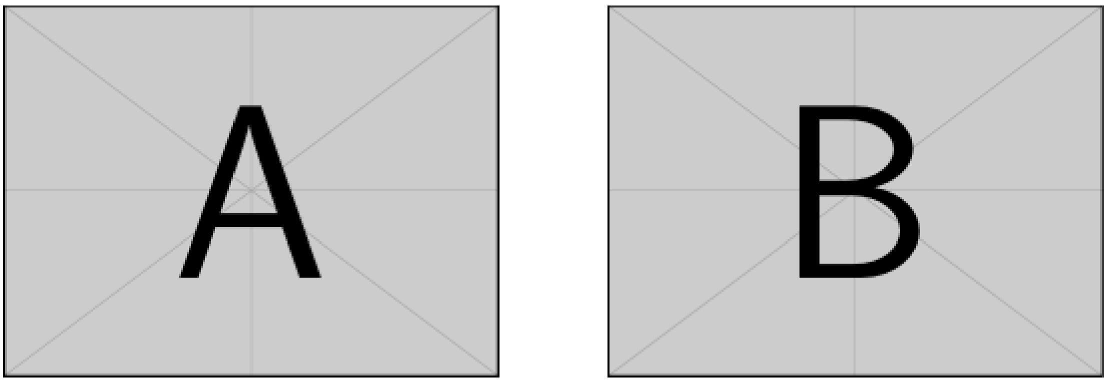
SL NO	USN	Student Name	MARKS		
			JAVA	DBMS	Python
1	4VM22IS001	ABC	89	60	90
2	4VM22IS002	MNQ	75	45	98
3	4VM22IS003	XYZ	67	55	59

Program 6

Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.

```
\documentclass{article}
\usepackage{graphicx} % For including graphics
\usepackage{subfigure} % For subfigures
\begin{document}
    \begin{figure}[htbp]
        \centering
        \subfigure[Subfigure A]{\includegraphics[width=0.45\textwidth]{example-image-a}}
        \hfill
        \subfigure[Subfigure B]{\includegraphics[width=0.45\textwidth]{example-image-b}}
        \caption{Side-by-side subfigures}
        \label{fig:side_by_side}
    \end{figure}
\end{document}
```

Output:



(a) Subfigure A (b) Subfigure B

Figure 1: Side-by-side subfigures

Program 7

Develop a LaTeX script to create a document that consists of the following two mathematical equations.

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2-4ac}}{2a} \\ &= \frac{-2 \pm \sqrt{2^2-4*(1)*(-8)}}{2*1} \\ &= \frac{-2 \pm \sqrt{4+32}}{2} \end{aligned}$$

$$\begin{aligned} \varphi^{\lambda}_{\sigma}A_t &= \sum_{\pi \in C_t} \operatorname{sgn}(\pi)\varphi^{\lambda}_{\sigma}\varphi^{\lambda}_{\pi} \\ &= \sum_{\tau \in C_{\sigma t}} \operatorname{sgn}(\sigma^{-1}\tau\sigma)\varphi^{\lambda}_{\sigma}\varphi^{\lambda}_{\sigma^{-1}\tau\sigma} \\ &= A_{\sigma t}\varphi^{\lambda}_{\sigma} \end{aligned}$$

```
\documentclass{article}
\begin{document}
    \begin{center}
            \Large{\textbf{Equations in LaTeX}}
    \end{center}

    \section*{Equation 1}
            \[
            x = \frac{-b \pm \sqrt{b^{2}-4ac}}{2a}
            \]

            \[
            = \frac{-2 \pm \sqrt{2^{2}-4*(1)*(-8)}}{2*1}
            \]

            \[
            = \frac{-2 \pm \sqrt{4+32}}{2}
            \]

    \section*{Equation 2}
            \[
            \varphi^{\lambda}_{\sigma}A_{t} = \sum_{\pi \in C_{t}} sgn(\pi)\varphi^{\lambda}_{\sigma}\varphi^{\lambda}_{\pi}
            \]

            \[
            = \sum_{\tau \in C_{\sigma t}} sgn(\sigma^{-1}\tau\sigma)\varphi^{\lambda}_{\sigma}\varphi^{\lambda}_{\sigma^{-1}\tau\sigma}
            \]

            \[
            = A_{\sigma t} \varphi^{\lambda}_{\sigma}
            \]
\end{document}
```

Output:

Equations in LaTeX

Equation 1

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{2^2 - 4 * (1) * (-8)}}{2 * 1}$$

$$= \frac{-2 \pm \sqrt{4 + 32}}{2}$$

Equation 2

$$\varphi_\sigma^\lambda A_t = \sum_{\pi \in C_t} sgn(\pi) \varphi_\sigma^\lambda \varphi_\pi^\lambda$$

$$= \sum_{\tau \in C_{\sigma t}} sgn(\sigma^1 \tau \sigma) \varphi_\sigma^\lambda \varphi_{\sigma^{-1}\tau\sigma}^\lambda$$

$$= A_{\sigma t} \varphi_\sigma^\lambda$$

Program 8

Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.

```
\documentclass{article}
\usepackage{amsthm} % For mathematical environments
\usepackage{lipsum} % Dummy text generator

% Define Theorem Styles
\newtheorem{theorem}{Theorem}[section]
\newtheorem{definition}[theorem]{Definition}
\newtheorem{corollary}[theorem]{Corollary}
\newtheorem{lemma}[theorem]{Lemma}
\newtheorem{example}[theorem]{Example}

\begin{document}

	\title{Presentation of Theorems, Definitions, Corollaries, and Lemmas}
	\author{Your Name}
	\date{\today}
	\maketitle

	\section{Theorems and Definitions}

	\begin{theorem}
		If \( a^2 + b^2 = c^2 \), then \( (a, b, c) \) is a Pythagorean triplet.
	\end{theorem}

	\begin{proof}
		Using the Pythagorean theorem, we can verify that for values such as \( (3, 4, 5) \), the
equation holds:
		\[
		3^2 + 4^2 = 9 + 16 = 25 = 5^2
		\]
		Thus, the statement is proven.
	\end{proof}

	\begin{definition}
		A **prime number** is a natural number greater than \(1\) that has no divisors other than
\(1\) and itself.
	\end{definition}

	\begin{example}
		For instance, the numbers \(2, 3, 5, 7\) are all prime numbers, while \(4\) and \(6\) are not
because they have additional divisors.
```

```
    \end{example}


\begin{corollary}
        If \( p \) is a prime number and \( p \) divides \( ab \), then \( p \) must divide either \( a \)
or \( b \).
    \end{corollary}

    \begin{lemma}
        If \( n \) is an even integer, then \( n^2 \) is also even.
    \end{lemma}

\end{document}
```

Output:

Presentation of Theorems, Definitions, Corollaries, and Lemmas

Your Name

June 19, 2025

1 Theorems and Definitions

Theorem 1.1. *If $a^2 + b^2 = c^2$, then (a, b, c) is a Pythagorean triplet.*

Proof. Using the Pythagorean theorem, we can verify that for values such as $(3, 4, 5)$, the equation holds:

$$3^2 + 4^2 = 9 + 16 = 25 = 5^2$$

Thus, the statement is proven. □

Definition 1.2. *A **prime number** is a natural number greater than 1 that has no divisors other than 1 and itself.*

Example 1.3. *For instance, the numbers 2, 3, 5, 7 are all prime numbers, while 4 and 6 are not because they have additional divisors.*

Corollary 1.4. *If p is a prime number and p divides ab, then p must divide either a or b.*

Lemma 1.5. *If n is an even integer, then n^2 is also even.*

Program 9

Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section

```
\documentclass{article}
\usepackage{lipsum}
\begin{document}
    \section{Introduction}
    \lipsum[1]
    This is the first paragraph with citations~\cite{citation1, citation2, citation3, citation4}.
    \lipsum[2]~\cite{citation5}
    This is the second paragraph with more citations~\cite{citation6, citation7, citation8, citation9,
citation10}.
    \begin{thebibliography}{99}
        \bibitem{citation1} Author1. Title1. Journal1. Year1.
        \bibitem{citation2} Author2. Title2. Journal2. Year2.
        \bibitem{citation3} Author3. Title3. Journal3. Year3.
        \bibitem{citation4} Author4. Title4. Journal4. Year4.
        \bibitem{citation5} Author5. Title5. Journal5. Year5.
        \bibitem{citation6} Author6. Title6. Journal6. Year6.
        \bibitem{citation7} Author7. Title7. Journal7. Year7.
        \bibitem{citation8} Author8. Title8. Journal8. Year8.
        \bibitem{citation9} Author9. Title9. Journal9. Year9.
        \bibitem{citation10} Author10. Title10. Journal10. Year10.
    \end{thebibliography}
\end{document}
```

Output:

1 Introduction

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. This is the first paragraph with citations [1, 2, 3, 4]. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris. [5] This is the second paragraph with more citations [6, 7, 8, 9, 10].

References

[1] Author1. Title1. Journal1. Year1.

[2] Author2. Title2. Journal2. Year2.

[3] Author3. Title3. Journal3. Year3.

[4] Author4. Title4. Journal4. Year4.

[5] Author5. Title5. Journal5. Year5.

[6] Author6. Title6. Journal6. Year6.

[7] Author7. Title7. Journal7. Year7.

[8] Author8. Title8. Journal8. Year8.

[9] Author9. Title9. Journal9. Year9.

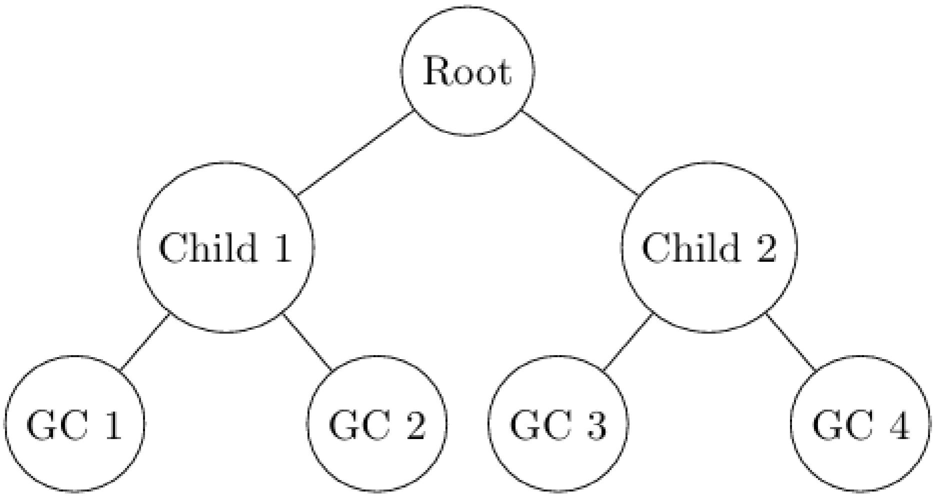
[10] Author10. Title10. Journal10. Year10.

Program 10

Develop a LaTeX script to design a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library

```
\documentclass{article}
\usepackage{tikz}
\begin{document}
    \begin{tikzpicture}[
        level 1/.style={sibling distance=4cm},
        level 2/.style={sibling distance=2.5cm},
        every node/.style={draw, circle, minimum size=1cm}
        ]
        \node {Root}
        child {node {Child 1}
        child {node {GC 1}}
        child {node {GC 2}}
        }
        child {node {Child 2}
        child {node {GC 3}}
        child {node {GC 4}}
        };
    \end{tikzpicture}
\end{document}
```

Output:


Root
Child 1
Child 2
GC 1
GC 2
GC 3
GC 4

Program 11

Develop a Latex Script to present an algorithm in the document using algorithm/algorithmic/ algorithm2e library.

```
\documentclass{article}
\usepackage{algorithm2e}
\begin{document}
\begin{algorithm}
    \caption{An algorithm to find Even or Odd}\label{alg:two}
    \KwData{$n \geq 0$}
    \KwResult{$y = x^n$}
    $y \gets 1$\;
    $X \gets x$\;
    $N \gets n$\;
    \While{$N \neq 0$}
    {
        \eIf{$N$ is even}
        {
            $X \gets X \times X$\;
            $N \gets \frac{N}{2}$;
        }
        {
            \If{$N$ is odd}
            {
                $y \gets y \times X$\;
                $N \gets N - 1$\;
            }
        }
    }
\end{algorithm}
\end{document}
```

Output:

Data: $n \geq 0$
Result: $y = x^n$
$y \leftarrow 1$;
$X \leftarrow x$;
$N \leftarrow n$;
while $N \neq 0$ **do**
    if N *is even* **then**
        $X \leftarrow X \times X$;
        $N \leftarrow \frac{N}{2}$;
    else
        if N *is odd* **then**
            $y \leftarrow y \times X$;
            $N \leftarrow N - 1$;
        end
    end
end

Algorithm 1: An algorithm to find Even or Odd

Program 12

Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice.

```
\documentclass{report}
\title{My Simple Report}
\author{Your Name}
\date{\today}
\begin{document}
    \maketitle
    \tableofcontents

    \chapter{Introduction}
    This is the introduction of my simple report.

    \section{About Project}
    Here, I'll provide some details of Introduction

    \chapter{Literature Review}
    This chapter describes the literature review used in my study.

    \chapter{Methodology}
    This chapter describes the methodology used in my study.

    \chapter{Results}
    Here are the results of my study.

    \chapter{Conclusion}
    In conclusion, I'll summarize my findings.
\end{document}
```

Output:

My Simple Report

Your Name

June 19, 2025

Contents

Chapter 1

Introduction

This is the introduction of my simple report.

1.1 About Project

Here, I'll provide some details of Introduction

Chapter 2

Literature Review

This chapter describes the literature review used in my study.

Chapter 3

Methodology

This chapter describes the methodology used in my study.

Chapter 4

Results

Here are the results of my study.

Chapter 5

Conclusion

In conclusion, I'll summarize my findings.